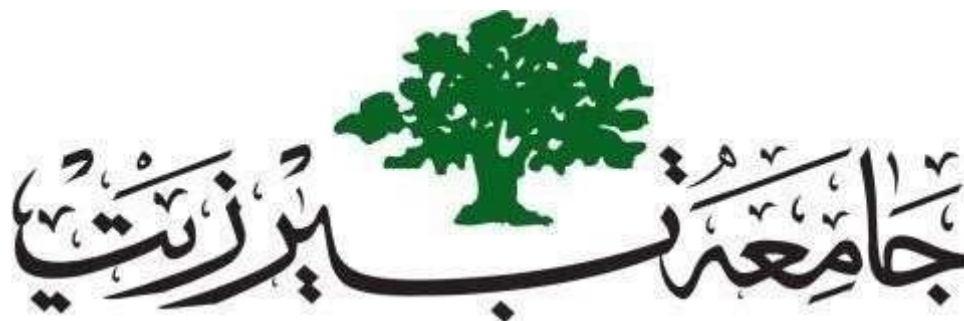




**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

Circuits and Electronics Laboratory - ENEE2103

PreLab Exp10: *The Operational Amplifier*

Prepared By:

Aya Dahbour 1201738

Instructor: Dr. Nasser Ismail.

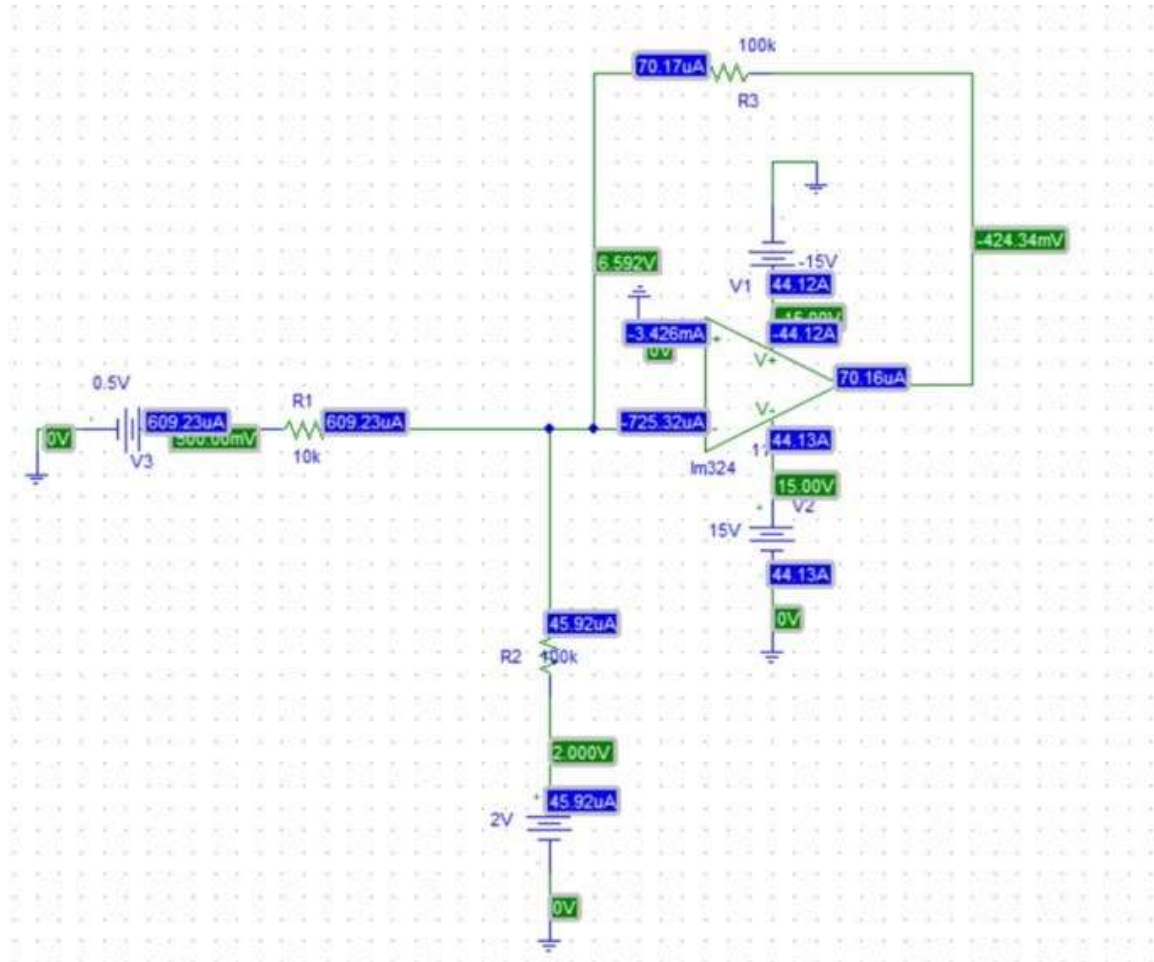
Assistant: Eng. Bilal Ismail

Section: 3

Date: 16/1/2023

I. Adding Application

Circuit connection , this connection is done for 3 values which are given in pre lab instructions



V1	V2	V _o
0.5	2	-6.592V
0.3	4	-6.592
-1.5	6	-5.844

$$I_1 + I_2 = I_0$$

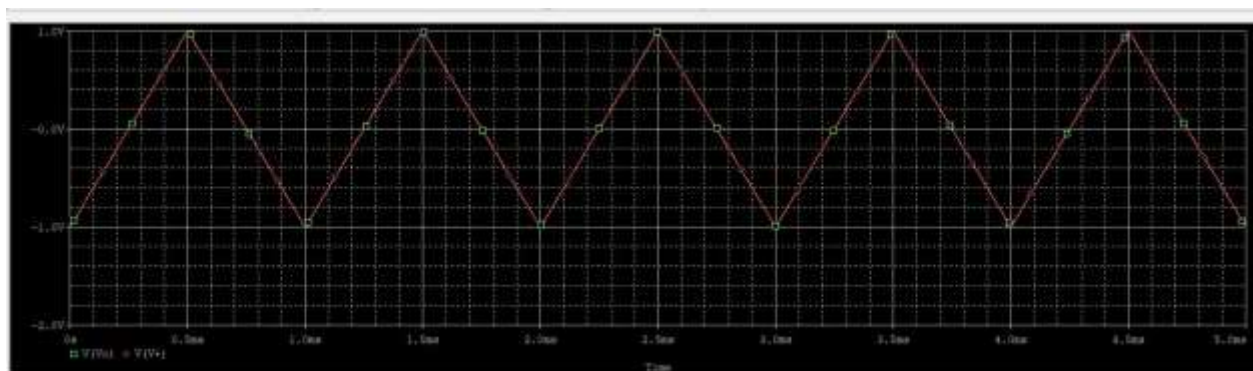
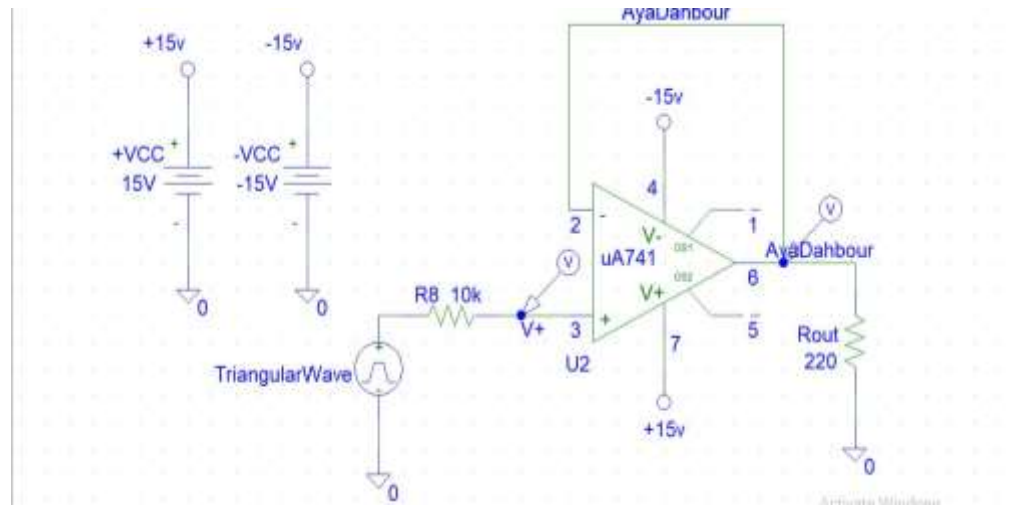
$$V_1/R_1 + V_2/R_2 = V_o/R_f$$

$$V_1 * R_f / R_1 + V_2 * R_f / R_2 = V_o$$

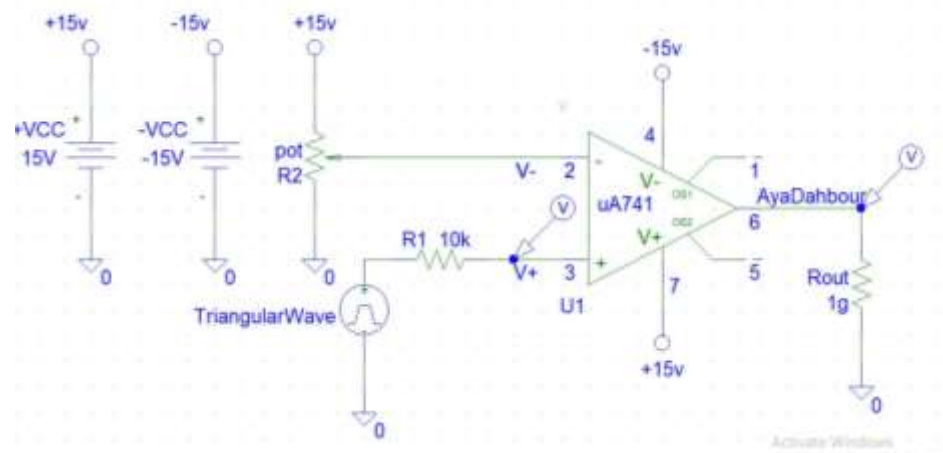
$$100\text{K}/10\text{K } V_1 + 100\text{K}/100\text{K } V_2 = V_o$$

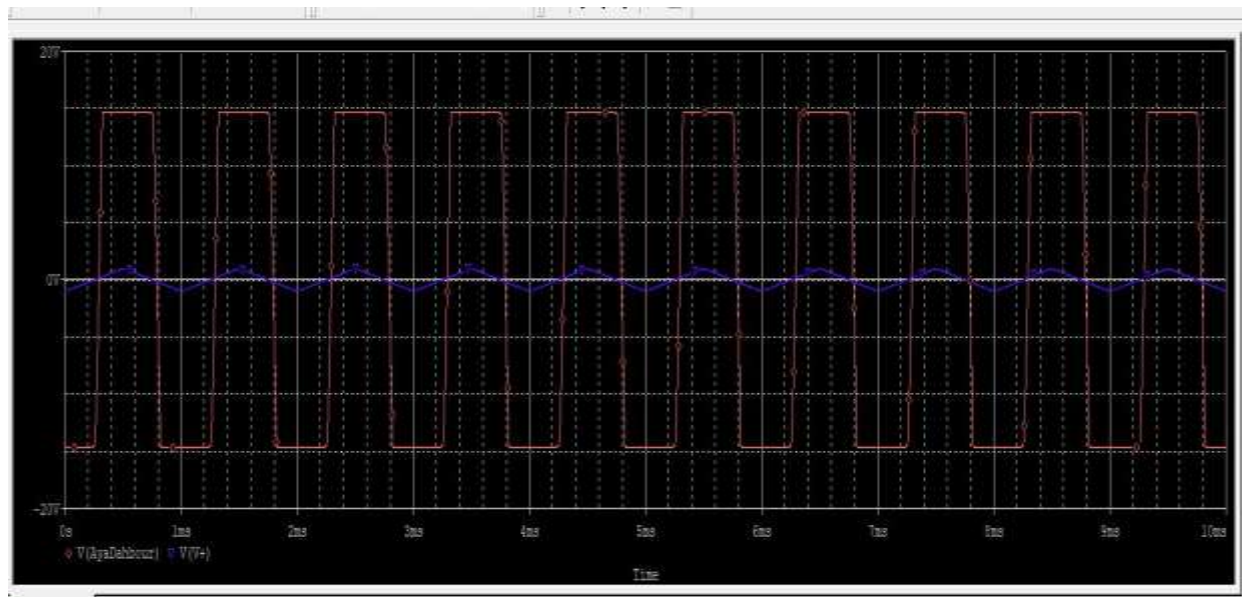
X = 10 , Y= 1 .

II. Voltage Follower Application



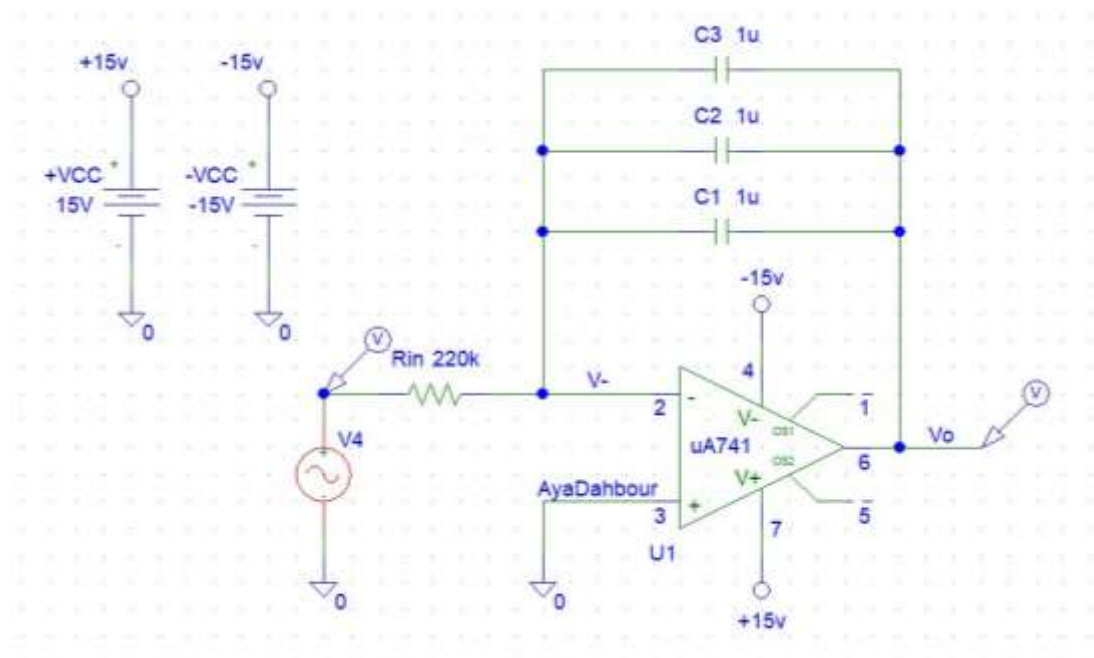
II. Comparator Application

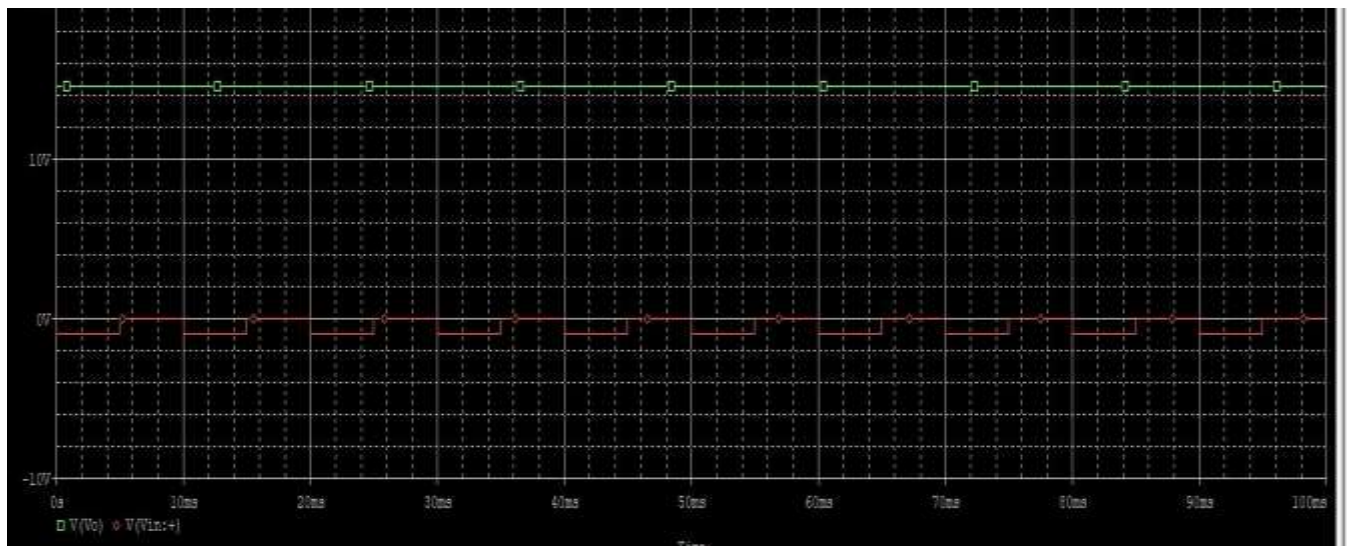




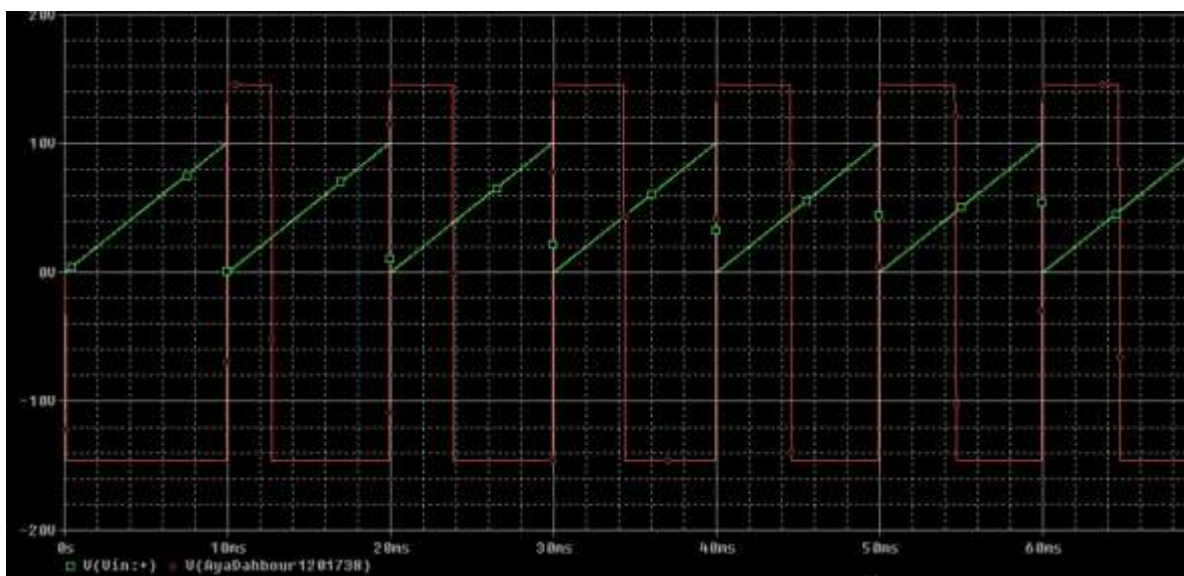
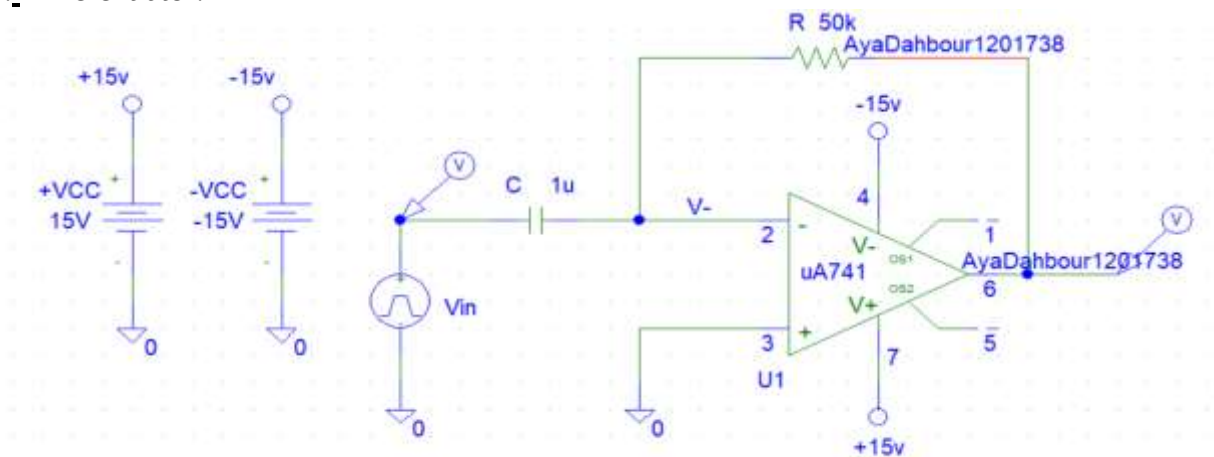
IV. Integrator and Differentiator

1. Integrator:





2. Differentiator:



V. To investigate the effect of adding hysteresis:

